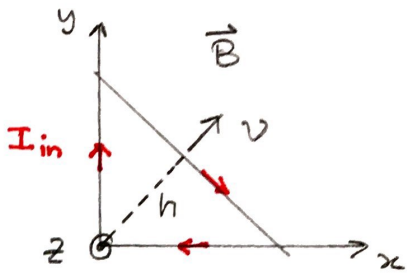
Final Exam Solution (00)

(1) $\vec{H} = \vec{a}_x - \vec{a}_y + \vec{a}_z \rightarrow \vec{B} = \mu_0 (\vec{a}_x - \vec{a}_y + \vec{a}_z)$ (Wb/m^2)

$\vec{a}_n = \vec{a}_z \rightarrow \vec{dS} = dx\,dy\,\vec{a}_z$

$$\phi = \mu_0 \int_0^y \int_0^x dx\,dy = \mu_0 h^2 \quad (Wb)$$



$$Emf = -d\phi/dt = -\mu_0 (2h \frac{dh}{dt})$$
$$= -2\mu_0 vh \quad (V)$$

$x = y = \sqrt{2}\,h$

$v = \frac{dh}{dt}$

$$I_{in} = \frac{Emf}{R} = -\mu_0 vh = -\mu_0 v^2 t$$

at $t = 10\,s \rightarrow |I_{in}| = 4\mu_0 \quad (mA)$

(2) $\vec{a}_n = \vec{a}_y \Rightarrow \vec{H}_{2n} = -\vec{a}_y \qquad \vec{H}_{2t} = \vec{a}_x + \vec{a}_z$

$\vec{B}_{2n} = -\mu_0 \vec{a}_y$

$\vec{B}_{1n} = \vec{B}_{2n} = -\mu_0 \vec{a}_y$

$\vec{H}_{1n} = \frac{\vec{B}_{1n}}{5\mu_0} = -\frac{1}{5}\vec{a}_y$

$\vec{H}_{1t} = \vec{H}_{2t} = \vec{a}_x + \vec{a}_z$

$\Rightarrow \vec{H}_1 = \vec{a}_x - \frac{1}{5}\vec{a}_y + \vec{a}_z \quad (A/m)$

(3) $\vec{E}(z,t) = 2\cos(\omega t - z)\,\vec{a}_x \Rightarrow \vec{E}_s(z) = 2e^{-jz}\,\vec{a}_x$

$\vec{D}_s(z) = 2(3\cdot3)\epsilon_0 e^{-jz}\,\vec{a}_x \Rightarrow \vec{J}_{ds}(z) = j\omega \vec{D}_s(z)$

$\vec{J}_{cs} = 0 \quad (\sigma = 0)$ $\qquad = j\omega(6.6)\epsilon_0 e^{-jz}\,\vec{a}_x$

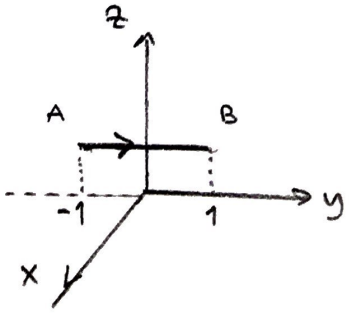
$$\vec{H}_s(z) = \frac{-\vec{\nabla} \times \vec{E}_s(z)}{j\omega\mu_0} = \frac{-\partial E_x/dz\,\vec{a}_y}{j\omega\mu_0} = \frac{2}{\omega\mu_0} e^{-jz}\,\vec{a}_y$$

$$\vec{H}(z,t) = \frac{2}{\omega\mu_0}\cos(\omega t - z)\,\vec{a}_y \quad (A/m)$$

(4) $\vec{B} = \mu_0(\vec{a}_x - \vec{a}_y + \vec{a}_z)$

$\vec{dl} = dy\,\vec{a}_y$

$$\vec{F}_m = \int I\,\vec{dl} \times \vec{B} = I \int_{-1}^{1} \mu_0 dy\,(\vec{a}_x - \vec{a}_z)$$



$|\vec{F}_m| = (0.1)\sqrt{2}\,\mu_0 \quad (N)$

$\vec{a}_{Fm} = (\vec{a}_x - \vec{a}_z)/\sqrt{2}$